

1. Let $f(n) = \sum_{y=1}^n (n^2 \cdot y^{13})$

Answer:

2. Let $n \geq 3$. Consider an ordered list of numbers, $L = (L_1, L_2, \dots, L_{n-1}, L_n)$

$$X_k = \sum_{i=2}^k (L_i - L_{i-1})$$

$$Y = \sum_{j=3}^n (X_j - X_{j-1})$$

Answer: